

CardioScan AI

GPU-Accelerated Cardiac Abnormality Detection from ECG Images using Deep Learning on NVIDIA H200

Team Name : HEARTNET

Team Members

RANGANATHA. M
20241IST0124

SHANTAYYA
20241IST0058



The Critical Challenge:

- Heart disease is the leading cause of death worldwide, claiming over 17 million lives annually
- Traditional ECG diagnosis is manual, time-consuming, and requires expert cardiologists
- No 24/7 continuous monitoring possible with current systems
- Inconsistent early detection leads to missed or delayed diagnoses

Project Objectives :

- Build a deep learning model to classify ECG signals into 5 cardiac categories
- Leverage NVIDIA H200 GPU for fast training and inference
- Apply transfer learning using ResNet-18 architecture
- Deploy as a web application for real-world accessibility

Dataset Used :

- MIT-BIH Arrhythmia ECG Dataset (Kaggle — shayanfazeli/heartbeat)
- 5 Classes: Normal, Supraventricular, Ventricular, Fusion, Unknown
- ECG signals converted into image format for CNN input
- 300 training images and 80 test images per class

Methodology:

- Step 1: Convert ECG signal CSV data into image format
- Step 2: Apply data augmentation (flip, rotation, color jitter)
- Step 3: Load pretrained ResNet-18 and apply transfer learning
- Step 4: Train on NVIDIA H200 GPU using PyTorch + CUDA

Model Architecture - ResNet-18 :

- Pretrained on ImageNet, fine-tuned on ECG images
- Custom classification head: Dropout → Linear(256) → ReLU → Dropout → Linear(5)
- Input size: 224 x 224 pixels – Optimizer: AdamW | Loss: CrossEntropy
- Batch size: 32 | Epochs: 15

NVIDIA GPU H200 Infrastructure :

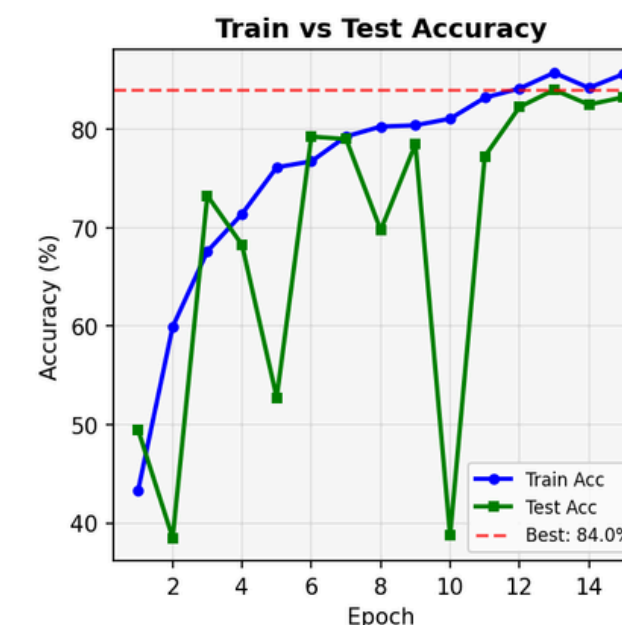
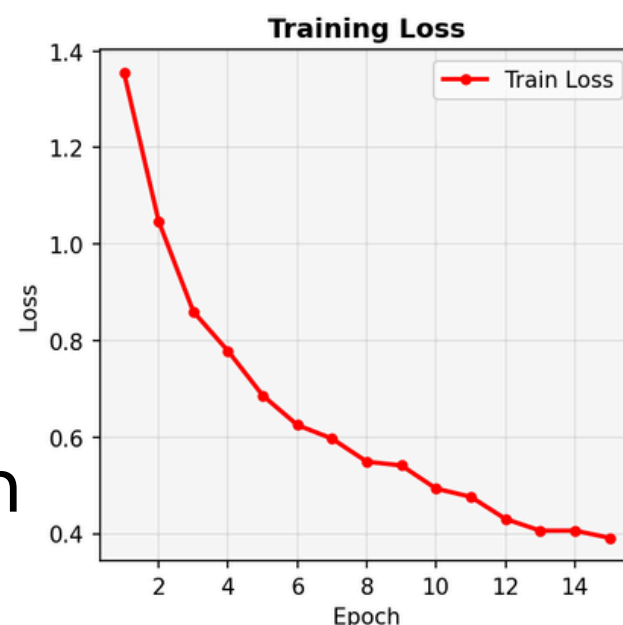
- GPU Memory: 141 GB HBM3e
- Memory Bandwidth: 4.8 TB/s
- FP8 Tensor TFLOPS: 3,958
- Architecture: NVIDIA Hopper (GH100)
- Used for: parallel batch processing, CUDA-accelerated training, fast inference, mixed precision (BF16)



Training Results :

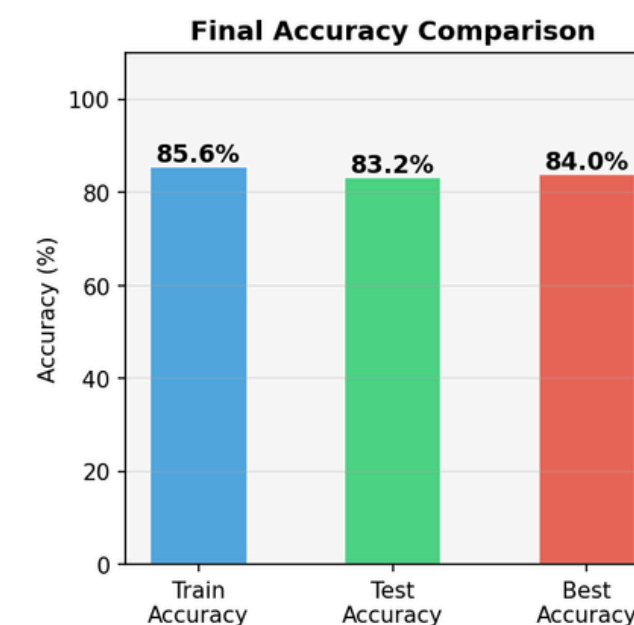
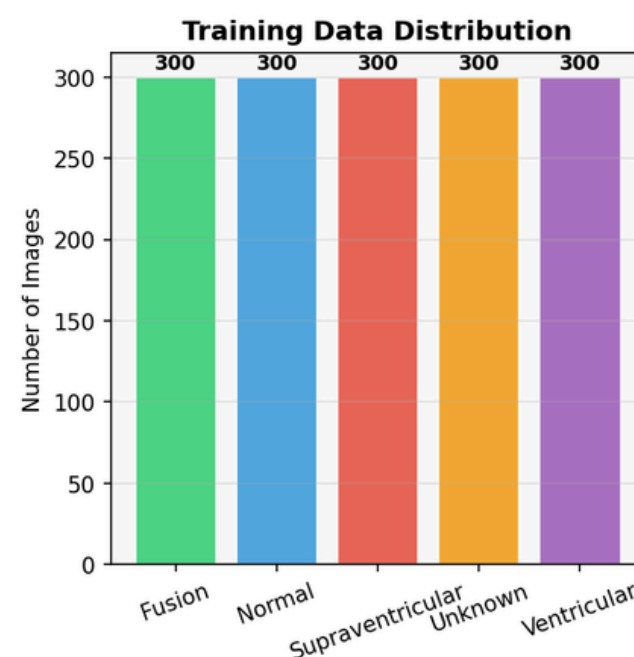
- Best Test Accuracy: 84%
- Train Accuracy: 85%
- Training Time: 1530.4 seconds on NVIDIA H200
- 2x speedup using BF16 mixed precision vs FP32

CardioScan AI — Training Results NVIDIA H200 GPU | ResNet-18 | ECG Classification



GPU Training Summary

Device:	CUDA
Architecture:	ResNet-18
Best Accuracy:	84.00%
Train Accuracy:	85.60%
Training Time:	1530.4s
Epochs:	15
Classes:	5
Batch Size:	32
Optimizer:	AdamW
Loss Fn:	CrossEntropy



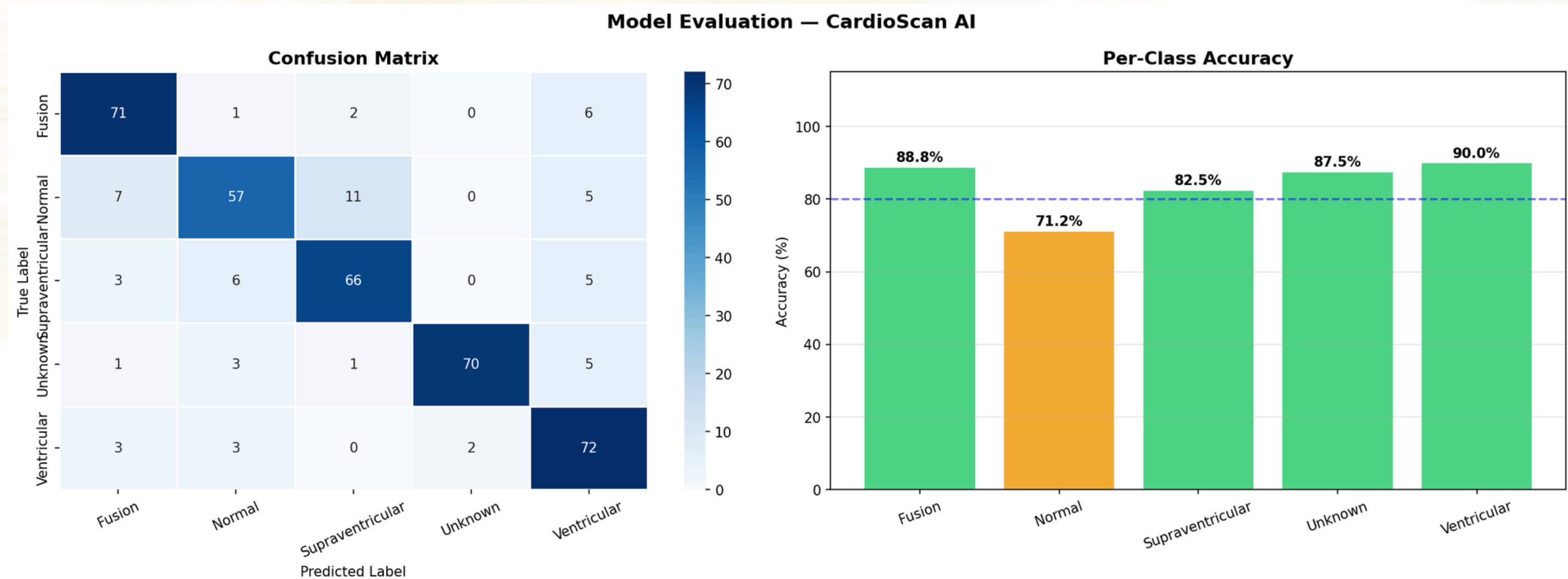
Project Information

Team:	HEARTNET
Project:	CardioScan AI
Institution:	Presidency University
Program:	AI & GPU Summer Internship 2026
Hardware:	NVIDIA H200 Tensor Core GPU
Dataset:	MIT-BIH ECG
Framework:	PyTorch + CUDA



Model Evaluation :

- Confusion matrix shows per-class prediction accuracy
- Normal class achieved highest accuracy
- Per-class accuracy comparison bar chart



Web Applications :

- Built using Flask + HTML/CSS/JS
- Pages: Home, About, Predict, Model Info, Contact
- Deployed on Render for public access
- GitHub: github.com/ranguuu143-cloud
- <https://nvidia-capstone-project-1.onrender.com>



Conclusion & Future Scope :

- GPU-accelerated CNN achieved high accuracy in ECG classification
- H200 enabled fast experimentation and real-time inference

Future Work:

- Train on larger real-world ECG datasets
- Add Explainable AI (Grad-CAM)
- Real-time continuous monitoring integration